UTMFOUNDATION_ FSPP 0034
TEST 2
18 MARCH 2021

ANSWER ALL QUESTIONS
DURATION: 1 HOUR

Write your name, IC No. and section on every page of your answer sheets.
Scan all your answers and upload the file in .pdf via Blackboard. Make sure you upload the correct file.

QUESTION 1 (10 MARKS)

(a) i) Sketch the pattern of magnetic field lines around all this magnet in Figure 1.

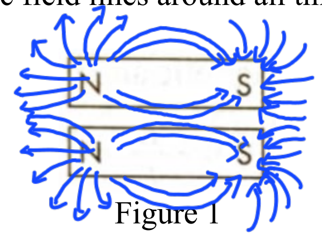


Figure 1

ii) Sketch the pattern of magnetic field lines around all this magnet in Figure 2.

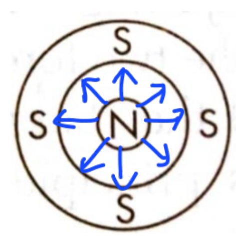


Figure 2

iii) Determine the direction of the magnetic force, ***F*** exerted on a charge as shown in Figure 3.

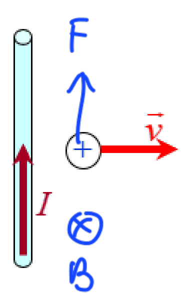


Figure 3

iv) Determine the direction of the magnetic force, ***F*** exerted on a charge placed in a magnetic field as shown in Figure 4.

fleming right hand

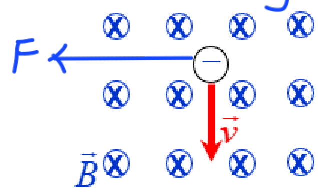


Figure 4

(2 marks)

(b) A straight horizontal rod with length 2000 cm and mass of 950 g is placed in a uniform magnetic field of 0.12 T perpendicular to the rod. The force acting on the rod just balances the rod's weight.

i) Sketch and label the diagram that shows the directions of the current, magnetic field, weight and force.

ii) Calculate the current in the straight horizontal rod. 3.88 A

(3 marks)

(c) Figure 5 shows the electron enters a uniform magnetic field, B = 12.78 T at a 45° angle $\vec{\boldsymbol{B}}$ with 500 eV of energy. Determine the radius of the electron's helical path. 4.18 x 10^-6 m

$B = 12.78\,T$

$\theta = 45^\circ$

$E = 8\times10^{-17}$

$r = ?$

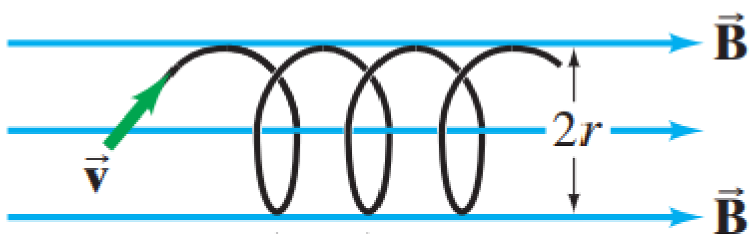


Figure 5

(5 marks)

$r = \frac{mv}{qB}$

$KE = \frac{1}{2}mv^2$

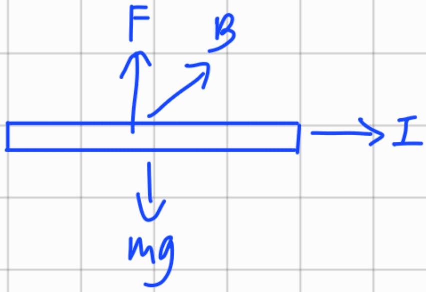
$8\times10^{-17} = \frac{1}{2}(9.11\times10^{-31})v^2$

$v = 1.325\times10^{7}\,ms^{-1}$

$r = \frac{(9.11\times10^{-31})(1.325\times10^{7})}{(1.6\times10^{-19})(12.78)} \times \sin 45^\circ$

$= 4.17\times10^{-6}\,m$

1) (b) (i)



(ii) $F = IlB\sin\theta$

$0.95(9.81) = I(20)(0.12)\sin 90$

$I = 3.88A$

(c) $500eV = 500 \times 1.6 \times 10^{-19}$

$= 8 \times 10^{-17} J$

2) (a) $d = 10 \times 10^{-3} m$, $I = 15A$

(i)

(ii) $B_A = \frac{\mu_0 I}{2\pi r} = B_B$

$= \frac{4\pi \times 10^{-7} \times 15}{2\pi(5 \times 10^{-3})}$

$= 6 \times 10^{-4}$ T into the page

$\Sigma B = B_A + B_B$

$= 1.2 \times 10^{-3}$ T into the page

(b) $l = 10m$, $d = 0.02m$, $I = 35A$

$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$

$F = 10\left(\frac{4\pi \times 10^{-7}(35)(35)}{2 \times \pi \times 0.02}\right)$

$= 0.1225 N$

$\therefore$ repulsive force

3) (b) $\frac{2.5m}{4} = 0.625m$

Area of square

$= 0.625^2$

$= 0.391 m^2$

$2\pi r = 2.5m$

$r = 0.398m$

Area of circle

$= \pi(0.398)^2$

$= 0.498 m^2$

$\mathcal{E} = B\cos\theta \frac{dA}{dt}$

$= 0.45(\cos 90°)\left(\frac{0.107}{5.5}\right)$

$= 8.75 \times 10^{-3} V$

$dA = 0.391 - 0.498 = |-0.107 m^2| = 0.107 m^2$

(c) $N_p = 350$, $N_s = 25$, $V = 100V$, $I = 0.9A$

$\frac{N_s}{N_p} = \frac{V_s}{V_p}$, $\frac{25}{350} = \frac{V_s}{100}$, $V_s = 7.14V$

QUESTION 2 (10 MARKS)

(a) Two long, straight wires X and Y are separated by a distance of 10 mm. The current in both wires is 15 A, but in opposite directions.

i) Draw the pattern of the magnetic field produced by the current carrying wires.

(2 marks)

ii) Find the magnitude and direction of the magnetic field at a point midway between the wires. 1.2 x 10^-3 T

(3 marks)

(b) Determine the magnitude and direction of the force between two parallel wires 10 m long and 2.0 cm apart, each carrying 35 A in the opposite direction. 0.12 N

(5 marks)

QUESTION 3 (10 MARKS)

(a) Determine the direction of current induced in the coil when bar magnet moves through the coil and then moves out from the coil as shown in Figure 6 (a) and Figure 6 (b), respectively.

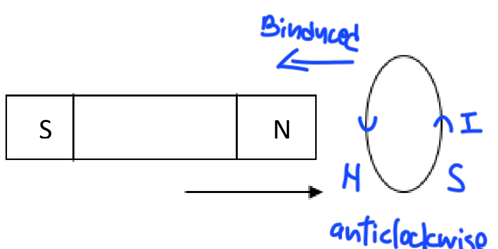


Figure 6 (a)

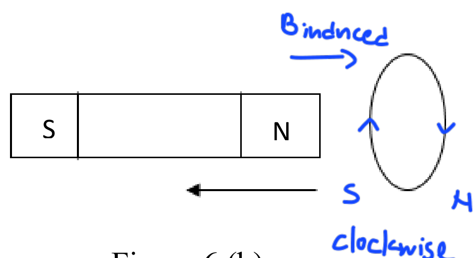


Figure 6 (b)

(2 marks)

(b) An emf is induced in a conducting loop of wire 2.5 m long as its shape is changed from circular to square. Determine the induced emf if the change in shape occurs in 5.5 s, and the uniform magnetic field of 0.45 T is perpendicular to the plane of the loop. 9.0 x 10^-3 V

(6 marks)

(c) A step-down transformer has 350 turns in the primary coil and 25 turns in secondary coil. The primary coil is connected to an AC power supply for 100 V and a current of 0.90 A is drawn when the power supply is turned on. Calculate the voltage across the secondary coil.

7.14 V

(2 marks)

APPENDIX

$e = -1.6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$g = 9.80$ m s^{-2}

$m_e = 9.11 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2N^{-1}m^{-2}$

$k = 9.0 \times 10^9$ N m^2 C^{-2}

1 eV= 1.60×10^{-19} J